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Elevated per- and polyfluoroalkyl substances (PFASs) in tap water and serum in a community near an abandoned paper mill

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Abstract

In 2018, elevated concentrations of per- and polyfluoroalkyl substances (PFASs) (1600 ng/L) were discovered in municipal wells, and the groundwater contamination was linked to an abandoned paper mill landfill in southwestern Michigan. From 2020–2021, we conducted a detailed exposure assessment in this community to provide insights about the mixture of PFASs from a paper mill source in drinking water and their persistence in human serum following long-term exposure. Water and serum were analyzed for >30 PFASs using LC-MS/MS. Approximately half of the study participants had private wells with lower PFAS concentrations (<220 ng/L) compared to municipal water (1600 ng/L), therefore, they were assigned as low (private well) and high (municipal well) exposure groups. The three predominant PFASs detected in the municipal well, perfluorooctanoic acid (PFOA) (670 ng/L), perfluorooctane sulfonic acid (PFOS) (740 ng/L), and perfluorohexane sulfonic acid (PFHxS) (19 ng/L), were also the most abundant in serum. Participants with high drinking water exposure had serum PFOA, PFOS and PFHxS

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Declaration of competing interest

The authors have the following declarations: CCC, CPH, and JLA have served as expert witnesses in PFAS litigation. All other authors have no declarations.

CRediT authorship contribution statement

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concentrations that were 15-fold, 5-fold, and 2-fold higher, respectively, compared to those in the low group. For each additional year participants reported drinking the municipal water, we observed a 6 %, 4 %, and 5 % increase in serum PFOA, PFOS and PFHxS, respectively. Overall, these results highlight the role of drinking water as a predominant source of PFAS exposure and the risk that abandoned landfills can pose to drinking water in nearby communities. This study also establishes a baseline for long-term toxicological impacts and evaluation of intervention effectiveness.

Keywords

PFAS; Serum; Water; Biomonitoring; Exposure

1. Introduction

Per- and polyfluoroalkyl substances (PFASs) are a group of chemicals used for their water, stain, and fire-resistant properties. They are used in many consumer products including clothing, carpets, non-stick cookware, electronics, food packaging, and aqueous firefighter foams (AFFF) used to fight flammable liquid fire (such as fuel fires), as well as many industrial processes including chrome plating and textile manufacturing (Sunderland et al., 2019; Tansel, 2022). PFASs are released into the environment via multiple pathways including industrial manufacturing of PFASs, use and training with AFFF, and landfills receiving PFAS-containing waste. Many PFASs are highly mobile and migrate easily through soils, surface and groundwater, air and dust, plants, and other organisms (Sunderland et al., 2019)

Many PFASs are persistent in the environment due to their strong carbon-fluorine bonds, and they also bioaccumulate in organisms and biomagnifying up the food chain (Tomy et al., 2004). Two common PFASs, perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) have been investigated extensively in animal and human studies, and found to affect multiple systems in the body including the liver, kidneys, thyroid, and immune system (ATSDR, 2021; Averina et al., 2021; Fenton et al., 2021; Grandjean et al., 2012, 2017; Kielsen et al., 2016; Sunderland et al., 2019). Although PFOA and PFOS began to be phased out of use in the early 2000s, they continue to persist in our environment. Meanwhile, they have been replaced with mixtures of other PFASs (Brase et al., 2021).

PFOA and PFOS have been detected in the blood of most Americans (Kato et al., 2011), with the highest levels among people occupationally exposed (Steenland and Woskie, 2012) and those with contaminated drinking water (Barton et al., 2020; Hall et al., 2023; Herrick et al., 2017). Drinking water is known to be an important exposure pathway for communities with PFAS contaminated water (Barton et al., 2020; Domingo and Nadal, 2019; Emmett et al., 2006; Hall et al., 2023; Hoffman et al., 2011; Vestergren and Cousins, 2009) whereas the predominant exposure source for the general population is believed to be diet followed by products and the indoor environment (De Silva et al., 2021; Ericson et al., 2008; Vestergren and Cousins, 2009). Serum PFOA and PFOS levels in the general population have been declining over the past decade following their phase out while increased monitoring has

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identified ground- and drinking water contamination linked to PFAS use and disposal(Coffin et al., 2023; Kato et al., 2011; Sunderland et al., 2019).

In June of 2018, elevated PFOA and PFOS (1410 ng/L total) was discovered in Parchment's municipal water supply wells serving a community of approximately 3,000 residents in southwestern Michigan (MPART, 2018). Of the 14 PFASs tested by the Michigan Department of Environment, Great Lakes, and Energy (EGLE), PFOA (670 ng/L) and PFOS (740 ng/L) were the predominant PFASs detected (88 % of 1600 ng/L) (MPART, 2018). A "Do not drink" advisory was issued and the municipal water was then connected to the Kalamazoo water system which had tested below the federal drinking water advisory of 70 ng/L for PFOS+PFOA (MPART, 2018). Many private wells in the area were also affected and those with high levels were eventually connected to the public water supply, while wells with lower concentrations continued to be used. The source of groundwater contamination was linked to a nearby former paper mill property and landfill, which had been used for the disposal of paper mill manufacturing waste (MPART, 2018). While many studies have investigated PFAS exposure via drinking water (Barton et al., 2020; Frisbee et al., 2009; Kotlarz et al., 2020), no studies have characterized exposure to the unique mixture of PFASs from a paper mill source.

Therefore, we conducted a detailed exposure assessment to provide useful insights about the mixture of PFASs from a paper mill source in drinking water and its persistence in human serum following long-term exposure. Specifically, we aimed to: 1) assess and quantify PFAS exposure in a community with PFAS-contaminated drinking water from a paper mill landfill three years after the water intervention and 2) quantify associations of PFASs in serum among participants with high (≤1600 ng/L) and lower (<220 ng/L) concentrations in drinking water.

2. Materials and Methods

2.1. Study Population

A community advisory board was formed comprised of local citizens and leaders, who provided input on the study design, survey, and recruitment materials. We recruited participants from August 2020 through November 2021. Individuals were eligible if they lived in the study area between 2005 and 2018 and reported having consumed the tap water. Our screening questionnaire identified 167 eligible participants, of which 129 (78 %) consented and enrolled in the study. Of these, 100 participants (77 households) scheduled home visits for collection of personal and environmental samples. We also administered detailed questionnaires to collect information on participant demographics, residential history, water source and filtration, and other data relevant to PFAS exposure. Each participant was visited at their home for collection of personal and environmental samples including blood and water, as described below. All protocols were approved by the Human Subjects Protection Program and Institutional Review Board at Michigan State University (STUDY00002925).

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2.2. Tap water

We collected unfiltered tap water from all participant homes currently using a private well as well as a subset of homes using public water. Drinking water was collected from an outdoor water spigot (95 % of homes) where possible, with the remainder of samples collected from the kitchen tap. All water sampling locations were flushed for at least one minute before collection. Our first round of sampling (2020–2021) collected approximately 7 mL of drinking water in a 15 mL polypropylene tube. Quality assurance samples collected included field blanks, trip blanks and 10 % duplicates. Samples from each home were double sealed in a polyethylene zip bag, transported on ice, stored at −20°C, and shipped on ice for PFAS analysis. The samples (42 homes) were analyzed for 48 PFASs using liquid chromatography-high resolution mass spectrometry (LC–HRMS) (Murray et al., 2019). Reporting limits (RL) were calculated for each analyte per sample by taking the maximum of the RL, trip blank, and field blank (Table S1). As RL's for PFASs in some samples were not sufficiently low due to unresolved contamination of field blanks, they were not used in our data analysis other than to estimate the maximum sum of all PFAS detected (ΣPFAS) in water samples (<220 ng/L) for homes using private wells (Table S2).

We conducted a second round of sampling in 2022 in which approximately 250 mL of drinking water was collected into a high-density polyethylene (HDPE) container. Quality assurance samples collected included field blanks, trip blanks and 20 % duplicates. Each sample and blank were double sealed in a polyethylene zip bag, stored at 4°C, and shipped within 96 h on ice for PFAS analysis. The samples (12 homes) were analyzed for 70 PFASs using liquid chromatography with tandem mass spectrometry (LC-MS/MS) and extracted using solid phase extraction as detailed by Pelch et al. (2023). Field and method blanks were below method detection limits (MDLs) for all PFASs. Laboratory control sample recoveries were within acceptable ranges and RL's and MDLs were at least ≤5 ng/L (Table S3).

2.3. Serum

In 2021, a certified medical technician performed a venous blood draw at each participant's home between June and October. Blood was collected into a plastic (polyethylene terephthalate) 10 mL red-top vacutainer. Each tube was allowed to clot at room temperature for 30 min after which it was transported in a cooler on ice. Within 5–8 h, samples were centrifuged at 1300 × g for 10 min at room temperature, and then the serum was aliquoted into 1 mL polypropylene cryovials using pipette tips triple rinsed with methanol, and frozen at −80°C. Samples were shipped on dry ice to the analytical laboratory for PFAS analysis.

Blood serum samples were extracted and analyzed for 35 PFASs using electrospray ionization LC-MS/MS as described by Hoxie et al. (2024). Analytes were quantified using a standard target approach with a 7-point calibration curve. A bovine matrix spike was conducted for QAQC purposes with all targeted analytes along with measurements in human serum Standard Reference Material (Hoxie et al. 2024). MDLs for individual analytes ranged from 0.01 to 0.4 ng/g (Table S4) and were set as three times the standard deviation of the laboratory processing blank values when detected in laboratory blanks or 10x the signal-to-noise ratio (0.010 ng). Serum PFAS concentrations are reported in μg/L (vs ng/L for water) per convention because they are typically found at higher concentrations.

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2.4. Reporting back participant results

Participants were given the option on their consent form to receive context-rich report back of overall results as well as a personalized report back of individual results For overall results, we generated summary statistics for tap water and serum PFAS concentrations. Tap water concentrations of PFASs were compared with PFAS concentrations from 2018 and Michigan's MCLs (MPART, 2020). Serum concentrations of PFASs were compared with the general adult U.S. population using the most recent results from the 2017–2018 National Health and Nutrition Examination Survey (NHANES)(CDC, 2023). We also generated a table and figure for each participant of their serum PFAS concentrations alongside concentrations for the overall study population (de-identified) and general population. Participants who provided a drinking water sample were provided a table and figure with the PFAS concentrations for their sample alongside concentrations first identified in the public water supply and Michigan's drinking water guidelines (MPART, 2020). We also created a letter with our contact information, an invitation to a virtual community meeting to discuss the results and answer any questions (including where to find the recording), select fact sheets, and a feedback survey. Report back materials were shared with the community advisory board and their feedback was incorporated. We mailed each participant a packet that included the letter, overall study results, their individual results, the fact sheets and the feedback survey.

2.5. Feedback Survey

We asked participants to complete a brief survey to provide feedback on the clarity, structure, and content of the report back. Participants were asked to indicate how well they agree with four statements, reporting on a scale of strongly disagree to strongly agree: (1) The report included information that was important to me; (2) Results are presented clearly; (3) The results are useful to inform my healthcare; and (4) The provided fact sheets and websites are helpful. Participants were also asked what they found most useful, what they did not like or had trouble with, and what other information would have been helpful to include. We generated summary statistics for feedback results.

2.6. Data Analysis

Our primary analysis applied multivariable linear regression to quantify associations of serum PFAS concentrations with exposure to the contaminated drinking water. Inclusion criteria for this analysis included serum PFAS data, PFAS water data estimates for pre-2018, and key demographic covariates including residential history, water source, sex, age, BMI, and education. Of the 100 participants with available serum PFAS data, 80 met these criteria: 18 were missing reliable estimates of PFAS water data and two were missing covariate data. Participants were divided into two exposure groups based on their residential history and water source prior to 2018. The high exposure group included participants who were on the municipal Parchment City water supply for at least 1 year between 2005 and 2018 ($n = 39$). The low exposure group included participants who were on a private well prior to 2018 and had their private well sampled during our study period ($n = 41$) (Fig. 1).

Serum PFAS concentrations <MDL were imputed with the MDL/√2 and duplicates were averaged (Hornung and Reed, 1990). Analysis of individual PFASs was restricted to those

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with a detection frequency ≥70 %: PFOS, PFOA, perfluoropentane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA), perfluoroheptane sulfonic acid (PFHpS), perfluorobutane sulfonic acid (PFBS), perfluorobutane sulfonic acid (PFPeA), N-methyl perfluorooctane sulfonamidoacetate (NMeFOSAA), and perfluorooctane sulfonamide (FOSAA). Σ9PFAS was calculated as the sum of these nine PFAS in serum. Σ5PFAS was calculated as the sum of the five of these reported for the general population and includes PFOS, PFOA, PFHxS, PFNA and PFHpS (CDC, 2023).

Results were tested for normality using the Shapiro Wilk test. As serum PFASs were not normally distributed, they were naturally log-transformed for use in regression models. Beta estimates were back-transformed to indicate the fold-difference. Potential confounders selected *a priori* for adjustment in the multivariable model included sex, age, body mass index (BMI), and education. Water covariates investigated included pre-intervention water filtration, consumption of bottled water, years on public water, and years worked in the study area. Geometric means were calculated for low and high exposure groups stratified by sex. We used a 2-sample test to assess for significant sex differences in serum PFAS. Adjusted statistical tests defined significance as $\alpha \geq 0.05$. All analyses were performed using Microsoft Excel and SAS 9.4.

3. Results

3.1. Study population

Participants included for this analysis ($n = 80$) were primarily adults (98 %) and most identified their race as white (97 %). More women (61 %) than men participated and most completed some college education or more (82 %) (Table 1). Most participants (>50 %) reported drinking unfiltered tap water (Table S6). This subset was similar to the full cohort ($n = 100$) with no significant differences in demographics (Table S5).

3.2. Water

In 2022, our second sampling event identified low concentrations in the current public water (< 8 ng/L PFOA+PFOS), dominated by perfluorobutane sulfonic acid (PFBS) (≤13 ng/L) and perfluoropropanoic (PFPrA) (≤26 ng/L). We also identified low concentrations in the private wells (≤12 ng/L PFOA+PFOS), with PFOA being less predominant (Table S7). We detected a total of 12 PFASs in both current public water and private wells, including 6 PFASs that were not included in the 2018 sampling (Fig. 2). Detection frequencies were 100 % for total PFOA and PFOS in current public water supply samples and 60 % for current private wells. PFPrA and perfluoropropane sulfonic acid (PFPrS) were the only PFASs that were detected in all current drinking water samples.

3.3. Blood

In our subset of 80 participants, twenty-seven of the thirty-five PFASs analyzed were detected in at least one or more samples (Table 2). Detection frequencies were 100 % for PFOS, PFOA, perfluoroheptane sulfonic acid (PFHpS), and perfluorohexane sulfonic acid (PFHxS). These were also the predominant PFAS in serum, with similar detection frequencies and geometric mean concentrations as the full cohort (Table S8).

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The low exposure group had similar concentrations of PFOA and PFOS compared to the general population and significantly lower medians for PFHxS, PFHpS, and perfluorononanoic acid (PFNA)(Fig. 3, Table S9). Participants with high drinking water exposure (municipal water) had significantly higher median concentrations of serum PFOA, PFOS, PFHxS, and PFHpS compared to those with low drinking water exposure (private wells) and the general population. Serum Σ9PFASs was 5-fold higher among participants in the high compared to low exposure group, adjusted for age, sex, BMI, and education (Table 3). The largest difference was observed with serum PFOA, where the high exposure group had 15-fold higher levels compared to the low exposure group. Serum PFOS, PFHpS, PFHxS, and, PFNA were also significantly higher among participants in the high exposure group. Effect estimates were similar with adjustment for water filtration, consumption of bottled water, or years worked in the study area (Table S10).

While sex was not a major predictor of serum levels in our multiple regression models, in a stratified analysis we observed significantly higher serum geometric means for PFOS, PFHpS and Σ9PFAS for males compared to females in the low exposure group (Table S11). This difference was not observed in the high exposure group.

3.3.1. Association between serum PFAS and years of tap water consumption:

To further understand the relationship between drinking water exposure and serum PFAS levels, we investigated the association between serum PFAS and years of tap water consumption. In the high exposure group, each additional year consuming the municipal water was associated with a significant 5 % increase in serum Σ9PFASs after adjusting for sex, BMI, and education (Table 4). This association was also observed for PFOS, PFOA, PFHxS, PFHpS, NMeFOSAA, and FOSAA. In the low exposure group, only PFHxS showed a significant association, with a 2 % increase in serum PFHxS for every additional year of private well water consumption, when adjusted for sex, BMI, and education. Effect estimates were similar when restricted to participants who reported their water consumption was 50 % or more tap water (Table S12).

3.4. Report back & feedback survey

Participants were mailed a packet including results, factsheets and a feedback survey. No participants opted out of receiving study results. Of the 129 participants, 52 participants completed and returned the feedback survey. We used four statements to collect quantitative feedback on the clarity, structure, and content of the overall report. >70 % agreed that the results were presented clearly, and the content was important and helpful to them. More than half (60 %) found the results useful to share with their healthcare provider, while several participants (29 %) responded that they were indifferent. Overall, responses were positive across the four statements (Fig. 4).

Participants were also given the opportunity to share what they liked and disliked about the report. Of the 52 participant responses, the predominant response (83 %) indicated appreciation of being informed of their results and especially their personal blood results with comparisons to national averages (Table S13). While most found the content and structure clear and understandable, a few individuals indicated they found it challenging to

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interpret. Suggestions for improvement included more information on PFASs in store-bought foods, government or location actions, or PFAS elimination and health effects. Overall, participants valued receiving their results and found the report to be useful.

4. Discussion

This is the first biomonitoring study for a community exposed to drinking water contaminated with PFASs from a paper mill source. Elevated concentrations of PFOA and PFOS in drinking water are similar to findings from a recent study in Norway which found elevated levels of PFOA and PFOS in surface water downstream from a paper mill (Langberg et al., 2021). In our community, the PFAS groundwater contamination had been linked to a nearby landfill that had received waste from a local paper mill that operated from 1909 to 2000 (Forist, 2005). This timing corresponds to the predominant use of two PFAS mixtures in specialty papers (e.g.,. food contact) that may explain high concentrations of both PFOA and PFOS in groundwater near the landfill (Fig. S1). The use of PFASs in food packaging began in 1967, after FDA approved the use of Zonyl RP, a paper fluoridizer that transforms to PFOA (Benesh, 2022; Hogue, 2021; IEHN, 2008), to make several industrial blends for use in food packaging and other products (Trier et al., 2011). A decade later in 1974, the PFOS precursor perfluorooctane sulfonamido ethanol phosphate diester (SAmPAP) began to be used in food contact paper and packaging (Benskin et al., 2012; Langberg et al., 2021). Shortly after closure of the local paper mill these PFASs (SAmPAP, PFOA, and PFOS) were phased out of use and replaced in paper products with many precursors that can transform to PFOA and other perfluorocarboxylic acids (PFCAs) such as polyfluoroalkyl phosphate esters (PAPs) and fluorotelomer alcohols (FTOH) (Coffin et al., 2023). Low detection of replacements such as PAPs in participant serum and drinking water is consistent with closure of the local paper mill just prior to that transition. However, monitoring for both legacy and current use PFASs should be conducted in other communities near paper mills and landfills, as both may be present due to the transition in use and high mobility in soil and water (Chow and Foo, 2023; Langberg et al., 2021). U.S. paper mills may stop using PFASs soon, as the U.S. Food and Drug Administration announced in February 2024 a voluntary market phase-out of PFAS-containing substances being sold by U.S. manufacturers of grease-proofing agents including for paper and paper-board packaging (CFSAN, 2024).

Our primary finding of the drinking water source as a strong predictor of serum PFASs is consistent with findings from other cohorts that have identified drinking water as a primary exposure pathway (Barton et al., 2020; Frisbee et al., 2009; Herrick et al., 2017; Kotlarz et al., 2020; MDHHS, 2024). Like our cohort, these studies also observed PFOA, PFOS, and PFHxS as the predominant PFASs in serum, even across different contamination sources (Fig. S2). For example, in a community with PFAS-contaminated drinking water due to local use of aqueous film-forming foam, PFHxS was the predominant PFAS in serum, followed by PFOS, and PFOA (Barton et al., 2020). Furthermore, a community near a PFOA manufacturing facility had predominantly PFOA and PFOS, though PFOS levels were comparable to the general population at that time (Frisbee et al., 2009). Similarly, in our study, the municipal wells had proportionally elevated levels of PFOA and PFOS and this was reflected in the serum for participants in the high exposure group.

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While sex was not a major predictor of serum levels in our multiple regression models, in a stratified analysis, we observed significantly higher serum geometric means for PFOS and PFHpS for males compared to females in the low exposure group. This is consistent with other studies and may be influenced by elimination via menstruation, childbirth, and breastfeeding (Calafat Antonia et al., 2007; Fromme et al., 2007). This difference was not observed in the high exposure group. Findings on sex differences in serum PFAS levels are mixed and can vary across studies. While some studies of highly exposed populations have observed lower serum concentrations in females compared to males (Ingelido et al., 2018; Siebenaler et al., 2017), others have not (Kannan et al., 2004; Olsen et al., 2004). These differing findings may be related to the compound and study population (Jain and Ducatman, 2019; Sonnenberg et al., 2023). While the proportion of non-menstruating women were similar between our low and high exposure groups, we observed a higher proportion of women in the high exposure group reported a hysterectomy (Table S14). Water filtration and bottled water use can affect an individual's exposure to drinking water contaminants, including PFASs (Herkert et al., 2020; Teymoorian et al., 2023). We did not find any associations between serum PFASs and water filtration and drinking water habits.

Overall findings from our results report-back feedback survey were consistent with previous reports from other studies including that participants appreciated being informed of their exposure, having their data compared to national averages, and that some had difficulty interpreting their data (Boronow et al., 2017; Ramirez-Andreotta et al., 2016a). This reinforces the importance of incorporating report-back of results in biomonitoring studies and indicates the need for continued improvement. For example, Ramirez-Andreotta et al. (2016b) reported that creating more opportunities for direct consultation may further improve participant understanding and environmental health literacy.

Our findings also highlight the importance of testing drinking water for PFASs, since the contamination was identified by Michigan's 2018 state-led monitoring program (MPART, 2018). This proactive monitoring enabled timely intervention to prevent further exposure and led Michigan to be among the few states to establish maximum contaminate levels (MCLs) in drinking water for 7 PFASs (MPART, 2020). U.S. EPA subsequently has established national enforceable maximum contaminant levels for six PFASs (EPA, 2024), which will promote more equitable identification and action on PFASs in drinking water systems nationwide.

Strengths of this study include utilization of sensitive analytical methods to investigate an extended panel of PFASs, including those used in paper food packaging, the wide range of PFAS concentrations in drinking water within our study population, and inclusion of a feedback survey for the results report-back. While this study occurred approximately 3 years after the intervention and captures serum levels that are likely lower than in 2018, results can serve as a baseline for future assessments and together with collection of other exposure data informs understanding of participants' exposure before the intervention in 2018. Future biomonitoring in this community could be used to assess changes in serum levels, quantify associations with health effects, and evaluate intervention strategies.

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5. Conclusions

Overall, these results highlight the role of drinking water as a predominant source of PFAS exposure for this community and the persistence of elevated concentrations in serum even 3 years after the intervention. The elevated levels of PFOA and PFOS detected in the municipal wells in 2018 may reflect the PFASs used at the paper mill before its closure in 2000 and its reported disposal of paper waste in the landfill near the municipal wells. This highlights the persistence of PFASs and the risk that abandoned landfills can pose to drinking water in nearby communities. These data establish a baseline for long-term toxicological impacts and evaluation of intervention effectiveness.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability

The data that has been used is confidential.

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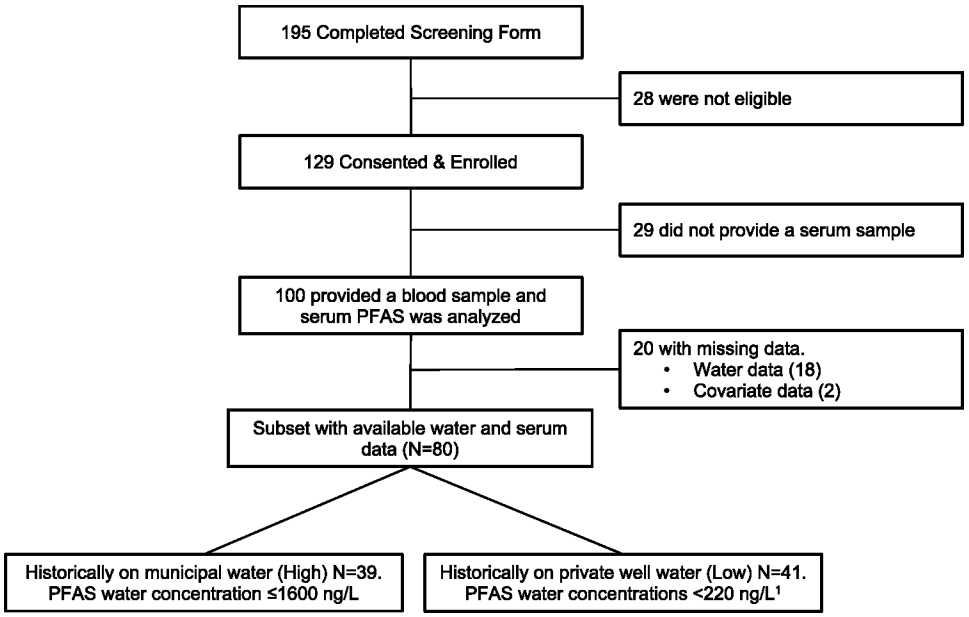


Fig. 1.

Selection Criteria.

[1] The maximum ΣPFAS concentration in private wells from the initial sampling event (2020–2021)

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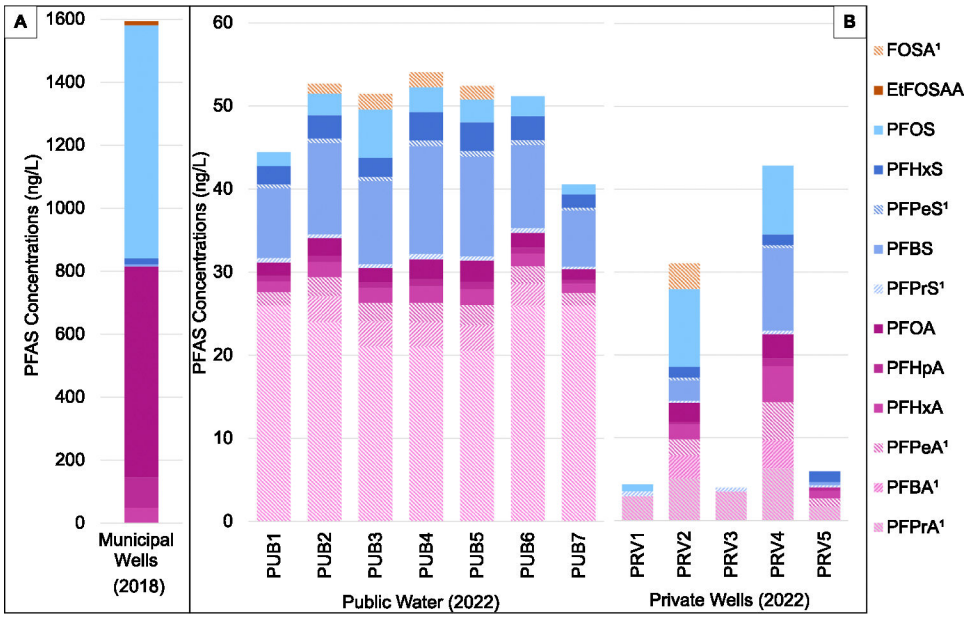


Fig. 2.

PFAS concentrations (ng/L) in the municipal wells at initial discovery (MPART, 2018) **(A)** as well as in tap water collected in our second sampling event (2022) **(B)**. Hatching indicates the six PFAS detected in our study[1] that were not investigated in the initial 2018 sampling During our second sampling event in 2022, we collected tap water samples from 12 homes (public, $n = 7$; private wells, $n = 5$). Each bar represents the PFAS water concentrations from an individual home. PUBn: public water sample from the nth home. PRVn: private well water sample from n^{th} home.

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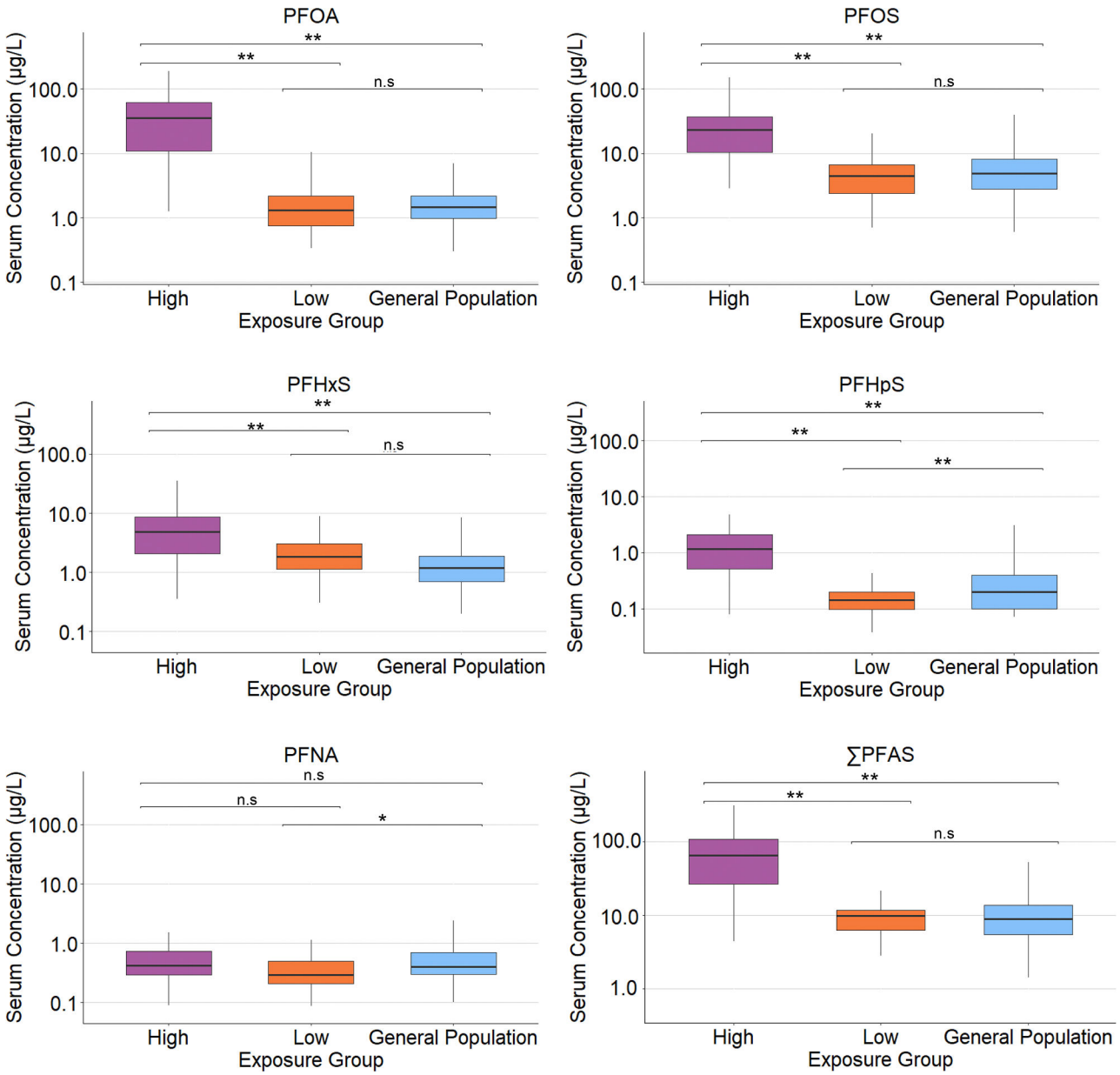


Fig. 3.

Median PFAS concentrations with interquartile range for adult participants in the high and low drinking water exposure groups compared to the U.S. general population. ΣPFAS, sum of PFOA, PFOS, PFHxS, PFHpS, and PFNA. High: Pre-2018 tap water contained ΣPFAS=1600 ng/L, $n = 38$ participants. Low: Pre-2018 tap water contained ΣPFAS <220 ng/L, $n = 40$ participants. General population: Weighted values from 2017 to 2018 NHANES, $n = 1{,}404$ participants (CDC, 2023). *$p < 0.05$, **$p < 0.01$, n.s.=not significant: $p > 0.05$ using Kruskal-Wallis test

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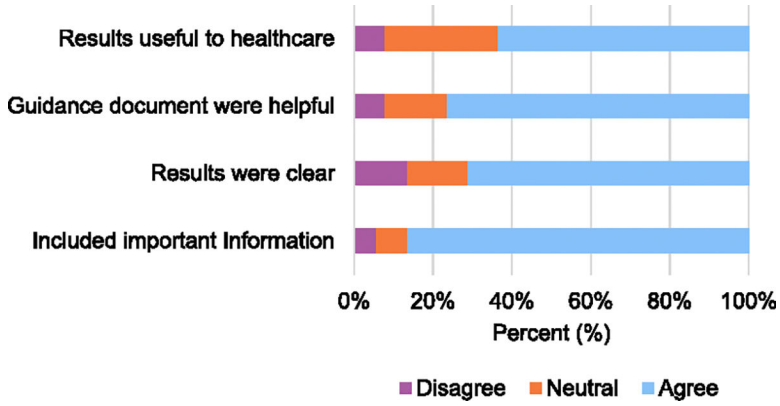


Fig. 4.

Response data from the results report-back feedback survey (N = 52). Participants were asked to indicate how well they agree with the four statements, reporting on a scale of strongly disagree to strongly agree: (1) The report included information that was important to me; (2) Results are presented clearly; (3) The results are useful to inform my healthcare; and (4) The provided fact sheets and websites are helpful.

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Table 1

Summary statistics for selected baseline characteristics of the subset population by exposure category.

Categorical Characteristics	**N (%)**		**Test Statistic[3] (p-value)**
	High Exposure (n = 39)[1]	**Low Exposure (n = 41)[2]**	
Sex			0.94 (0.33)
Females	26 (67)	23 (56)	
Males	13 (33)	18 (44)	
Age			2.03 (0.36)
Minor (< 18)	1 (3)	1 (2)	
Adult (18 – 65)	31 (79)	27 (66)	
Senior (>65)	7 (18)	13 (32)	
Race			4.0 (0.14)
White	39 (100)	38 (95)	
Black	0	0	
Asian	0	0	
Multiracial	0	2 (5)	
BMI			2.04 (0.36)
Normal	10 (27)	8 (20)	
Overweight	12 (31)	19 (46)	
Obese	17 (44)	14 (34)	
Education Level			
High School Graduate or lower	12 (31)	3 (7)	9.19 (0.03)
Some College or Technical School	14 (36)	13 (32)	
College 4 years or more	9 (23)	17 (41)	
Graduate or Professional Degree	4 (10)	8 (20)	
Continuous Characteristics	**Mean (SD)**		**Test Statistic[4] (p-value)**
Age	53.13 (14.96)	53.32 (16.68)	−0.05 (0.96)
BMI	31.35 (9.48)	29.09 (6.10)	1.26 (0.21)

[1] Missing Race (n = 4), BMI (n = 3), Education (n = 3);

[2] Missing Race (n = 1);

[3] Statisical comparisons for categorical variables were tested using Chi^2 test.

[4] Statistical comparisons for continuous variables were tested using Welch's t-test.

SD: Standard deviation.

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Table 2

Serum PFAS (μg/L) summary statistics for the subset (n = 80).

Analyte	MDL	DF (%)	50th Percentile	95th Percentile	Maximum
PFBA	0.236	3	<MDL	<MDL	0.47
PFPeA	0.121	90	0.36	0.83	1.25
PFHxA	0.107	1	<MDL	<MDL	0.36
PFHpA	0.114	5	<MDL	0.02	0.48
PFOA	0.063	100	4.2	86.22	187.82
PFNA	0.072	99	0.34	1.41	4.56
PFDA	0.062	59	0.07	0.34	1.47
PFUnA	0.023	68	0.03	0.17	0.79
PFDoA	0.014	15	<MDL	0.04	0.08
PFBS	0.233	74	0.71	2.37	4.0
PFPeS	0.008	55	0.01	0.1	0.31
PFHxS	0.037	100	2.77	18.58	35.54
PFHpS	0.015	100	0.34	3.2	4.74
PFOS	NA	100	7.11	58.62	149.22
br-PFOS	0.028	100	3.38	34.29	51.18
n-PFOS	0.019	100	4.09	27.27	107.55
PFNS	0.008	1	<MDL	<MDL	0.01
PFDS	0.018	3	<MDL	<MDL	0.03
PFDoS	0.014	0	-	-	-
GenX	0.080	0	-	-	-
4:2 FTS	0.409	0	-	-	-
6:2 FTS	0.371	0	-	-	-
5:3 FTCA	0.014	16	<MDL	0.04	0.06
7:3 FTCA	0.014	1	<MDL	<MDL	0.15
6:2 FTCA	0.014	0	-	-	-
8:2 FTCA	0.014	0	-	-	-
FBSA	0.014	33	<MDL	0.01	0.10
FHxSA	0.014	44	<MDL	0.02	0.13
FOSAA	0.005	84	0.01	0.1	0.28
NMeFOSAA	0.005	99	0.08	0.56	1.32
NEtFOSAA	0.006	35	<MDL	0.04	0.11
6:2 PAP	0.959	0	-	-	-
8:2 PAP	0.740	0	-	-	-
6:2 diPAP	0.015	4	<MDL	<MDL	0.06
8:2 diPAP	0.014	4	<MDL	<MDL	0.47
6:2/8:2 diPAP	0.014	1	<MDL	<MDL	0.15

NA, Not applicable. - Not calculated. MDL, method detection limit.

DF: Detection frequency, Percent of samples detected above the MDL.

PFOS: sum of br-PFOS and n-PFOS.

br: branched.

n: linear.

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Table 3

Geometric mean serum PFAS concentrations (μg/L) and fold-difference for high and low drinking water exposure groups

Analyte	Geometric Mean (95 % CI)		Linear Regression Fold-Difference (95 %CI)	
	High	Low	Crude	Adjusted
PFOS	19.41 (14.15, 26.62)	4.30 (3.34, 5.53)	4.52 (3.04, 6.71) *	4.85 (3.00, 6.65) *
PFOA	24.27 (16.01, 36.81)	1.40 (1.06, 1.83)	17.40 (10.72, 28.23) *	15.60 (9.63, 25.26) *
PFHxS	4.50 (3.21, 6.32)	1.91 (1.50, 2.44)	2.36 (1.56, 3.55) *	2.15 (1.49, 3.41) *
PFNA	0.42 (0.33, 0.53)	0.32 (0.25, 0.40)	1.31 (0.93, 1.84)	1.55 (1.13, 2.12) *
PFHpS	0.97 (0.70, 1.34)	0.14 (0.11, 0.18)	6.74 (4.53, 10.01) *	6.56 (4.42, 9.75) *
PFBS	0.69 (0.52, 0.93)	0.72 (0.55, 0.94)	0.96 (0.65, 1.42)	0.96 (0.63, 1.46)
PFPeA	0.37 (0.31, 0.45)	0.29 (0.24, 0.35)	1.27 (0.98, 1.66)	1.09 (0.83, 1.42)
NMeFOSAA	0.06 (0.04, 0.09)	0.10 (0.07, 0.14)	0.62 (0.35, 1.08)	0.55 (0.31, 0.97) *
FOSAA	0.01 (0.01, 0.02)	0.02 (0.01, 0.02)	0.74 (0.47, 1.16)	0.67 (0.43, 1.06)
^9PFAS	54.87 (39.49, 76.23)	10.74 (8.90, 12.96)	5.11 (3.54, 7.38) *	5.10 (3.54, 7.35) *

CI: Confidence intervals.

High: Pre-2018 tap water contained ΣPFAS=1600 ng/L, $n = 39$.

Low: Pre-2018 tap water contained ΣPFAS <220 ng/L, $n = 41$.

Crude: Univariate linear regression.

Adjusted: Multiple linear regression adjusted for age, sex, education, and body mass index.

* $p < 0.05$.

Σ9PFAS, sum of PFOS, PFOA, PFHxS, PFNA, PFHpS, PFBS, PFPeA, NMeFOSAA, and FOSAA.

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Table 4

Fold-difference (95 % CI) in serum PFASs for every year spent drinking the pre- 2018 drinking water.

Analyte	High		Low	
	Fold-difference (95 % CI)		Fold-difference (95 % CI)	
	Crude	Adjusted	Crude	Adjusted
PFOS	1.04 (1.02, 1.07) *	1.04 (1.01, 1.07) *	1.01 (0.99, 1.03)	1.01 (0.99, 1.03)
PFOA	1.07 (1.03, 1.10) *	1.06 (1.03, 1.10) *	1.01 (0.99, 1.03)	1.01 (0.99, 1.03)
PFHxS	1.05 (1.02, 1.08) *	1.05 (1.02, 1.07) *	1.02 (1.00, 1.04) *	1.02 (1.00, 1.04) *
PFNA	1.02 (1.00, 1.04)	1.02 (1.00, 1.04)	1.00 (0.98, 1.02)	1.01 (0.99, 1.03)
PFHpS	1.05 (1.02, 1.07) *	1.04 (1.02, 1.07) *	1.00 (0.99, 1.02)	1.01 (0.99, 1.03)
PFBS	0.99 (0.97, 1.02)	0.99 (0.96, 1.02)	1.00 (0.98, 1.02)	1.00 (0.98, 1.02)
PFPeA	1.01 (0.99, 1.02)	1.00 (0.98, 1.02)	1.00 (0.98, 1.01)	1.00 (0.98, 1.00)
NMeFOSAA	1.05 (1.01, 1.08) *	1.04 (1.00, 1.08) *	1.02 (0.99, 1.05)	1.02 (0.99, 1.05)
FOSAA	1.04 (1.01, 1.06) *	1.03 (1.01, 1.06) *	1.01 (0.99, 1.04)	1.01 (0.99, 1.04)
Σ9PFAS	1.05 (1.02, 1.07) *	1.04 (1.02, 1.07) *	1.01 (0.99, 1.02)	1.01 (1.00, 1.02)

CI: Confidence intervals.

High: Pre-2018 tap water contained ΣPFAS=1600 ng/L, n = 39 participants.

Low: Pre-2018 tap water contained ΣPFAS <220 ng/L, n = 41 participants.

Crude: Univariate linear regression.

Adjusted: Multiple linear regression adjusted for sex, education, and body mass index.

* $p < 0.05$.

Σ9PFAS, sum of PFOS, PFOA, PFHxS, PFNA, PFHpS, PFBS, PFPeA, NMeFOSAA, and FOSAA.